

Attention Is All You Need

Vaswani et al. (2017)

Problem & contribution. Recurrent sequence-to-sequence models propagate state one position at a time, which serializes training within each example and makes distant tokens interact through long computational paths. The paper introduces the *Transformer*: an encoder-decoder model that uses attention for all cross-position communication and applies the same small feed-forward transformation independently at every position. It therefore processes all training positions in parallel while letting any position directly select information from any other; the decoder remains autoregressive at generation time.

Routing mechanism. For query vectors Q , key vectors K , values V , and key width d_k , scaled dot-product attention is

$$\text{Attn}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V.$$

The query-key scores measure which positions match; the scale factor keeps their variance near one when components have unit variance, preventing large logits from saturating softmax; row-wise normalization turns each query row into weights; and the weighted value mixture retrieves the selected information. Thus each head acts as differentiable content-addressed memory: queries are lookups, keys are addresses, and values are payloads. Encoder self-attention derives Q, K, V from the same sequence; masked decoder self-attention does likewise but blocks future keys; cross-attention derives Q from the decoder and K, V from encoder outputs. One mixture can blur different relations, so h heads learn separate projections:

$$H_i = \text{Attn}(QW_i^Q, KW_i^K, VW_i^V), \\ \text{MHA} = \text{Concat}_{i=1}^h(H_i)W^O.$$

Each head uses reduced dimensions, so several routing patterns cost roughly one full-width head rather than h full models.

Architecture. Each of six encoder layers applies self-attention then a position-wise feed-forward network. Each of six decoder layers applies masked self-attention, encoder-decoder attention, then the feed-forward network. The decoder receives the previously emitted symbols shifted right; a triangular mask assigns future-token logits $-\infty$, preserving the autoregressive factorization, while cross-attention lets each generated position query the encoded input. A tied embedding/output matrix maps final decoder states to vocabulary logits. Every sublayer uses the residual form $\text{LayerNorm}(x + \text{Sublayer}(x))$. The feed-forward map $\max(0, xW_1 + b_1)W_2 + b_2$ is shared across positions but not layers. Because attention itself is order-agnostic, fixed sinusoidal coordinates are added to embeddings (scaled by $\sqrt{d_{\text{model}}}$): $PE_{p,2i} = \sin(p/10000^{2i/d})$ and $PE_{p,2i+1} = \cos(p/10000^{2i/d})$. A fixed offset can be represented as a linear transformation of these codes, giving the model a usable representation of relative displacement; learned positions performed almost identically in the reported ablation. The base model uses $d_{\text{model}} = 512$, $d_{\text{ff}} = 2048$, $h = 8$, and $d_k = d_v = 64$. Training minimizes teacher-forced next-token cross-entropy with label smoothing 0.1. Adam uses

$\beta_1 = 0.9$, $\beta_2 = 0.98$, and $\epsilon = 10^{-9}$ with learning rate $d_{\text{model}}^{-1/2} \min(t^{-1/2}, t^{-3/2})$; linear warmup stabilizes early optimization, then inverse-square-root decay lowers the step size. Residual and attention dropout are 0.1 in the base model.

Its computational tradeoff helps explain the design:

Layer	Work	Sequential	Max path
Self-attention	$O(n^2d)$	$O(1)$	$O(1)$
Recurrent	$O(nd^2)$	$O(n)$	$O(n)$
Convolution	$O(knd^2)$	$O(1)$	$O(\log_k n)$

Dense attention was cheaper than recurrence when $n < d$ in the translation setting and exposed short gradient paths, but its quadratic sequence term is the central scaling weakness. Multi-head projections preserve this routing advantage while allowing different representation subspaces and relative positions to be selected simultaneously.

Evidence & scope. On WMT 2014 English-German (about 4.5M sentence pairs), the 65M-parameter base model reports 27.3 BLEU at an estimated 3.3×10^{18} training FLOPs after 100k steps (12 hours on eight P100 GPUs); the 213M-parameter big model reports 28.4 BLEU, over 2 BLEU above prior reported systems including ensembles, after 300k steps (3.5 days). The big variant also doubles model width to 1024, doubles heads to 16, doubles feed-forward width to 4096, and raises dropout to 0.3, so its gain does not isolate one change. For English-French (36M pairs), the abstract and table report 41.8 BLEU while the prose says 41.0, an internal source inconsistency. The reported translations average recent checkpoints and use a shared byte-pair vocabulary, beam size 4, and length penalty 0.6; these conditions demonstrate strong quality/compute tradeoffs in these tasks, not architecture-independent superiority. Development ablations support multiple heads without showing that more is always better: one head scored 24.9 BLEU, 8/16 heads 25.8, and 32 heads 25.4. Label smoothing worsened perplexity but improved accuracy and BLEU, warning that likelihood and task metrics need not rank models identically. A four-layer Transformer reached 91.3 F1 on WSJ-only constituency parsing and 92.7 with semi-supervision, evidence of transfer beyond translation rather than proof of universal superiority.

Limits. Dense attention trades parallel training and short paths for $O(n^2)$ memory and work; the paper leaves local attention for long inputs to future work. Autoregressive decoding still generates tokens sequentially, and the experiments do not report matched end-to-end inference latency or memory. Sinusoidal extrapolation beyond trained lengths is a motivation, not an established result. Attention plots show syntactic-looking and long-range patterns, but qualitative examples do not prove that attention weights faithfully explain predictions. The results cover translation and parsing under substantial compute; architecture ablations are development-set point estimates, not uncertainty-aware general scaling laws or controlled comparisons against recurrent models at equal compute.